

◆ 1. Basics of Spring Boot

1. What is Spring Boot?

Spring Boot is a framework built on top of Spring that simplifies application development by providing:

- Auto-configuration
 - Embedded servers
 - Production-ready features
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2. Why use Spring Boot instead of Spring?

- No XML configuration
 - Embedded servers (Tomcat, Jetty)
 - Faster development
 - Auto configuration
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3. What are the main features of Spring Boot?

- Auto Configuration
 - Spring Boot Starters
 - Embedded Server
 - Actuator
 - Spring Initializr
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4. What is Spring Boot Starter?

Predefined dependency bundles like:

- `spring-boot-starter-web`
 - `spring-boot-starter-data-jpa`
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5. What is auto-configuration?

Spring Boot automatically configures beans based on dependencies in the classpath.

◆ 2. Core Concepts

6. What is Dependency Injection?

A design pattern where objects are injected instead of created manually.

7. What are different types of DI?

- Constructor Injection ☒ (preferred)
 - Setter Injection
 - Field Injection
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8. What is @SpringBootApplication?

Combination of:

- @Configuration
 - @EnableAutoConfiguration
 - @ComponentScan
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9. What is @ComponentScan?

Scans and registers beans automatically from the package.

10. Difference between @Component, @Service, @Repository, @Controller?

- @Component → Generic
 - @Service → Business logic
 - @Repository → DB layer (exception handling)
 - @Controller → Web layer
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◆ 3. REST API Development

11. What is @RestController?

Combines:

- `@Controller`
 - `@ResponseBody`
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12. Difference between `@Controller` and `@RestController`?

- `@Controller` → returns views
 - `@RestController` → returns JSON
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13. What is `@RequestMapping`?

Maps HTTP requests to handler methods.

14. Difference between `@GetMapping`, `@PostMapping`, etc.?

Shortcuts for HTTP methods:

- GET, POST, PUT, DELETE
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15. What is `@PathVariable` vs `@RequestParam`?

- `@PathVariable` → from URL path
 - `@RequestParam` → from query parameters
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◆ 4. Spring Data JPA

16. What is Spring Data JPA?

Simplifies database operations using JPA and repositories.

17. What is `JpaRepository`?

Provides built-in CRUD methods.

18. Difference between `CrudRepository` and `JpaRepository`?

- JpaRepository extends CrudRepository
 - Adds pagination & sorting
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19. What is @Entity?

Marks a class as a database table.

20. What is @Id?

Defines the primary key.

21. What is Hibernate?

ORM framework used internally by Spring Boot.

◆ **5. Database & Transactions**

22. What is @Transactional?

Manages database transactions automatically.

23. What is Lazy vs Eager loading?

- Lazy → loads when needed
 - Eager → loads immediately
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24. What is DTO?

Data Transfer Object used to transfer data between layers.

◆ **6. Exception Handling**

25. What is `@ControllerAdvice`?

Global exception handler.

26. What is `@ExceptionHandler`?

Handles specific exceptions.

◆ 7. Configuration & Properties

27. What is `application.properties`?

Used to configure app settings.

28. Difference between `application.properties` and `application.yml`?

- `.properties` → key=value
 - `.yml` → hierarchical structure
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29. What is `@Value`?

Injects values from properties file.

30. What is `@ConfigurationProperties`?

Maps entire config to a class.

◆ 8. Spring Boot Actuator

31. What is Actuator?

Provides production-ready features like:

- Health check

- Metrics
 - Monitoring
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◆ 9. Security

32. What is Spring Security?

Framework for authentication & authorization.

33. What is JWT?

JSON Web Token used for secure authentication.

◆ 10. Microservices & Advanced

34. What is Microservices architecture?

Breaking app into small independent services.

35. What is Eureka Server?

Service discovery tool.

36. What is Feign Client?

Used for service-to-service communication.

37. What is API Gateway?

Single entry point for all services.

38. What is @Bean?

Defines a bean manually in configuration.

39. What is @Autowired?

Injects dependencies automatically.

40. What is Spring Boot DevTools?

Provides hot reload for development.

11. Scenario-Based Questions

41. How do you handle exceptions globally?

Using `@ControllerAdvice`

42. How do you connect to a database?

Using:

- `application.properties`
 - Spring Data JPA
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43. How do you secure APIs?

Using:

- Spring Security
 - JWT
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44. How do you improve performance?

- Caching
- Lazy loading
- Pagination

45. How do you debug Spring Boot app?

- Logs
- Actuator
- Debug mode